# User Guide: AUTOSAR HLD Document Analysis Assistant

## What you need

A Linux computer with Docker and Docker Compose, Ollama, and about 16 GB of RAM. No GPU is needed or used. The first container build needs an internet connection.

## Start the system

1. In a terminal, start the model server and leave it open: `CUDA_VISIBLE_DEVICES=-1 ollama serve`. The first time only, download the model with `ollama pull qwen2.5:3b`.
2. In a second terminal, go to the `Code` folder and run `docker compose up -d --build`. The first build takes several minutes.
3. Wait about 30 seconds, then open `http://localhost:8501` in a browser.

To stop the system, run `docker compose down` in the `Code` folder.

## Sign in

Enter the API key for your role in the sidebar. The demo keys are in `Code/store/users.json`. A viewer can ask and read, an engineer can also review and edit, and an admin can also read the audit log. Then pick a document in the sidebar.

## The tabs

- **Ask:** type a question and choose a document version. The answer shows its status, its source (database lookup or language model), a confidence label, and citations with section and page.
- **Components and ports:** the extracted rows for the selected document. Engineers can change a row's review status or its values and press Save.
- **Findings:** the results of the six consistency checks, with a review status for each.
- **Functional flow:** for each component, its purpose from the document plus what it reads, what it writes, and which component provides or uses each signal.

- **Export:** download the extracted data as JSON or CSV. Rejected rows are left out.
- **Audit log:** every action with time and user (admins only).

## Reading an answer

| Status | Meaning |
|---|---|
| answered | The answer came from the database, or the model answered and its names and numbers were found in the cited text |
| not found | The model found nothing in the evidence |
| out_of_scope | The question is not covered by the approved documents, and no model was used |
| unverified | The model's answer could not be checked against its citations. Do not rely on it |

The confidence label reflects how well the search matched, not whether the answer is correct. Always check the cited page, and remember that a human must review any output before it is used for design, compliance or release.

## Example questions

- Which component provides the brake status? (version 1.0 and 1.1 give different answers)
- What is the period of the cruise control runnable?
- What changed in version 1.1?
- What are the inconsistencies in this design?
- What does SWC_DisplayCluster do?

## Troubleshooting

- **Cannot reach the API:** run `docker compose ps` in the `Code` folder and check that both containers are up.
- **Error about the local LLM server:** Ollama is not running. Start it as in step 1.

- **A port is already in use:** stop any API or web app you started by hand, then run `docker compose down` and `docker compose up -d`.
- **Open questions take 20 to 60 seconds:** the model runs on the CPU. Questions answered from the database come back instantly.
- **After reviewing rows:** the graph and the flow reports are built from the rows at build time, so rebuild them in the notebook to include your changes.